TwinHealth AI

A Digital Twin that warns of glucose spikes 2 hours ahead in type 2 diabetes (India)













Our Team: cosmic bloom

DEEPIKA E

JAYASRI D

DHATCHAYANI S

J BHAGYALAKSHMI

College : GRT Institute of Engineering and Technology



Problem statement and use case

- Type 2 diabetes is very common in India, but doctors see patients only in short, infrequent visits.
- Glucose spikes after meals happen between visits and go unnoticed.
- Health records and wearable data sit in different places and are never combined.

Our use case

Warn 2 hours ahead when glucose is likely to rise above 180 mg/dL, show why, and let the doctor test what-if changes.

The decision it supports

Which patient should I look at today, and should I review meals, sleep or activity?



Our solution: a virtual patient

01.

Virtual patient

Keeps 24 hours of state.

Updates with every new 5-minute reading.

Cleans bad or missing input.

02.

Fused prediction

Health record (old data) + wearable and glucose sensor (live data).

Output: spike risk and a 2-hour glucose curve.

03.

Safe alerts

Alerts only when risk stays high and data is trustworthy.

Otherwise the twin says “cannot judge” instead of guessing.





Technology stack

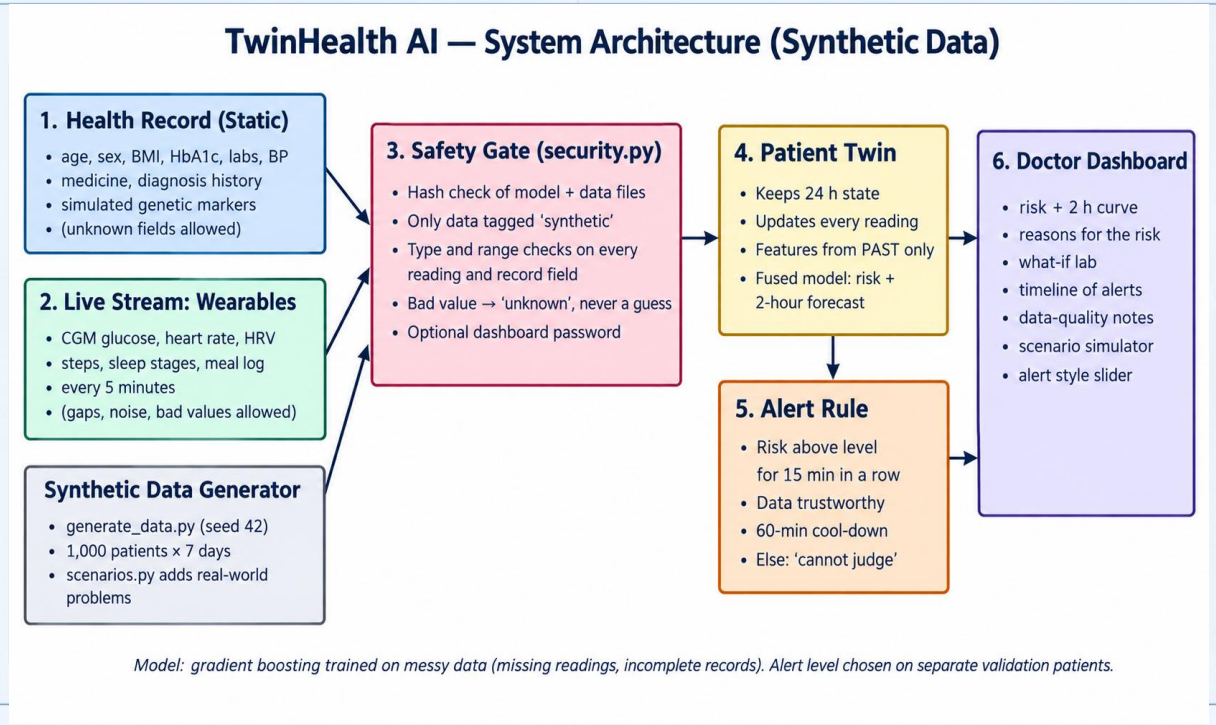
Layer	Technology	Purpose
Programming language	Python	Core application and ML logic
Data processing	Pandas, NumPy	Data cleaning, time-series processing, EHR/sensor fusion
Machine learning	Scikit-learn	Classification, preprocessing, model evaluation
Model storage	Joblib	Save and load trained model bundles
Dashboard	Streamlit	Interactive doctor-facing interface
Visualizations	Matplotlib	Glucose curves, forecast comparisons, explanation charts
Data storage	CSV files	Synthetic EHR and sensor datasets for the prototype
Security utilities	Python security module, Streamlit session state	Model integrity checks and demo access control
Version control	Git and GitHub	Source code, documentation and submission
Testing	Pytest	Data handling, predictions and Digital Twin behavior

Security utilities are not, by themselves, evidence of full healthcare regulatory compliance.





System architecture







Data: two synthetic streams

Static stream: health record

- Age, sex, BMI, diagnosis, family history
- HbA1c, fasting glucose, cholesterol, blood pressure
- Medicine, BP and cholesterol history
- Simulated genetic risk score and TCF7L2-style alleles

Live stream: wearable and sensor

- Glucose (CGM), heart rate, HRV, steps
- Sleep stages and last night’s sleep
- Meal log with Indian dishes (carbs, GI class)
- Every 5 minutes; gaps and noise added

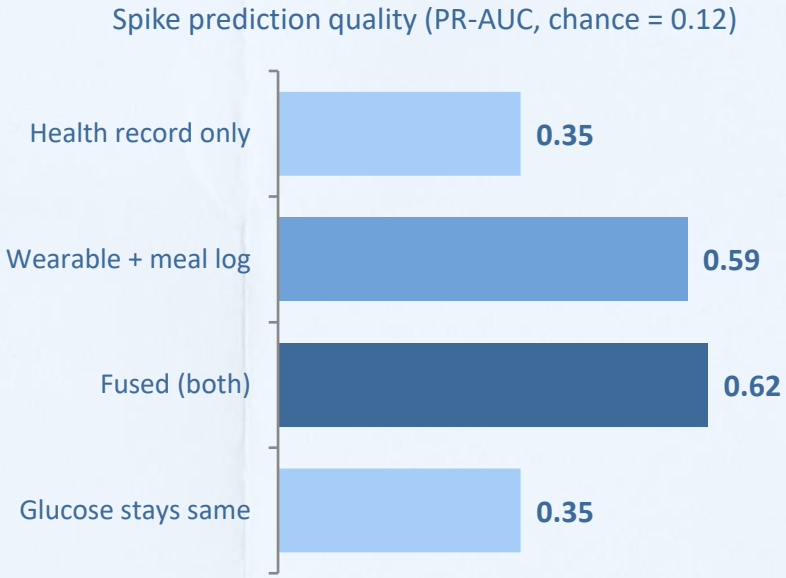
1,000 patients × 7 days × 288 readings a day | spike rate 10.5% | 800 train, 200 unseen | every row tagged synthetic

AI/ML models and frameworks

Component	Suggested model or method	Framework
Two-hour spike-risk prediction	Logistic Regression or Random Forest classifier	Scikit-learn
Glucose forecasting	Persistence baseline plus a regression model for future glucose values	NumPy / Scikit-learn
EHR and sensor fusion	Feature-level fusion of historical and recent sensor features	Pandas / Scikit-learn
Model explanations	Feature-group perturbation or ablation analysis	Python / Scikit-learn
Digital Twin state	Stateful Python class updated by incoming readings	Python
Model evaluation	PR-AUC, precision, recall, false-alert rate and forecast MAE	Scikit-learn / NumPy

Framework = the software used to build a model. Algorithm = the model itself.

Model and proof that fusion helps



- Gradient boosting, trained on messy data (gaps, incomplete records).
- Features use only past readings; a test proves no future leakage.
- Tested on unseen patients; alert level chosen on separate patients.
- Removing the health record lowers accuracy: both streams matter.



The twin and its alert rule

- Keeps 24 h of memory and updates per reading
- Rejects absurd values (glucose 8 or 650) and bad timestamps
- Handles late, duplicate and missing readings
- Explains risk by hiding one input group at a time
- What-if runs on a copy, never on the real twin

An alert fires only if ALL hold

1. Risk at or above the alert level
2. High for 3 readings in a row (15 min)
3. Glucose not already above 180
4. Data is trustworthy
5. No alert in the last 60 minutes



Doctor dashboard

Risk and 2-hour curve

Live glucose, forecast, danger line at 180, and a clear High / Moderate / Low badge.

Reasons

Which inputs raise or lower the risk: glucose trend, last meal, steps, sleep, health record.

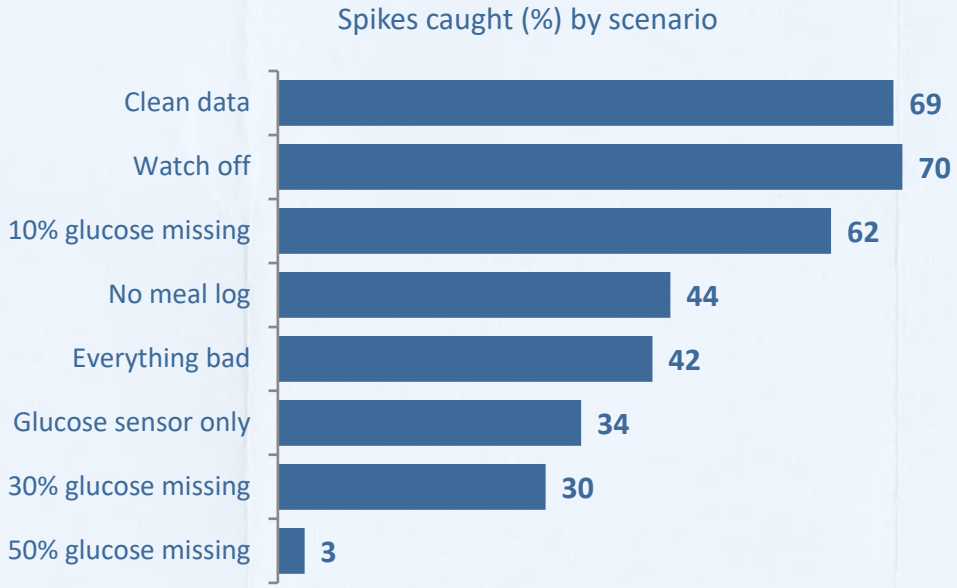
What-if lab

Try more sleep, a short walk or a lighter meal. Shows how the model reacts, not proof of treatment.

Problem simulator

Break sensors or hide parts of the record live. Alert-style slider trades false alarms for missed spikes.

What if the data is messy?



- With much missing glucose the twin says “cannot judge” (60% of the time at 30% missing, 94% at 50%) instead of guessing.
- No lab results: warning time falls from 35 to 20 minutes.
- 23 scenarios and 19 patient groups tested; 43 automatic tests pass.

Results at the low-false-alarm setting

69%	65%	0.69	35 min
of spikes caught	of alerts were right	false alerts per patient-day	median warning time

- 100 unseen synthetic patients, clean data. No setting gives zero false alarms and every spike: fewer false alarms means more missed spikes.
- On separate validation patients the same level gave 0.40 false alerts a day, so results vary with the patient sample.
- The doctor chooses: Low false alarms, Balanced, or Catch more spikes.





Security and privacy by design

Synthetic data only

Loaders refuse anything not tagged synthetic.

Hash-checked files

A swapped model or data file is refused.

Input cleaning

Every reading and record is type- and range-checked.

Private dashboard

Local only, optional password, XSRF on, no tracking.

Pinned, scanned libraries

pip-audit clean; bandit no medium or high issues.

Honest limits

No software promises zero breaches. See SECURITY.md.







Limits and real-data plan

Limits and weak spots

- All data is simulated by rules we wrote: scores show the pipeline works, not clinical accuracy.
- Prediabetes: only 50% of spikes caught.
- Sensor reading 20 mg/dL too high: score falls from 0.64 to 0.43.
- Not simulated: kidney disease, insulin use, pregnancy.

Real-data test (ShanghaiT2DM)

- 109 recordings, 112,462 readings. Chinese patients, many on insulin (not simulated).
- PR-AUC: twin 0.35, glucose stays same 0.33, straight-line trend 0.41, chance 0.15.
- **Honest finding:** the twin does NOT beat a simple trend rule on real data.
- Next: tune on real patients, fairly split.



Conclusion

01. A working digital twin that fuses a health record with live sensor data and warns 2 hours ahead.

02. Built to cope with missing data and to say “cannot judge” instead of guessing.

03. Honest results: false alarms and weak spots are measured and shown.

04. Next: test on real open data, extend to more conditions, validate with Indian cohorts and clinicians.

Link: https://github.com/deepikae1504/twinhealthai.git



Thank You











